

1(a)	force proportional to product of masses and inversely proportional to square of separation	B1
	idea of (gravitational) force between point masses	B1
1(b)	gravitational force provides/is the centripetal force	B1
	$GM/R^2 = R\omega^2$ or $GM/R^2 = v^2/R$	M1
	$\omega = 2\pi/T$ or $v = 2\pi R/T$	M1
	algebra leading to $R^3/T^2 = GM/4\pi^2$	A1
1(c)	$(6.67 \times 10^{-11} \times M)/4\pi^2 = (4.38 \times 10^6)^3 / (2.44 \times 3600)^2$	C1
	$M = 6.45 \times 10^{23} \text{ kg}$	A1